

KLH UNIVERSITY

Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms. Md Razia
Student Name	Adunuri Bhavana Sri
Roll Number	2520030232
Branch	CSE
Course Outcome	2

Case Study Topic: HealthNet – Intelligent Hospital Resource Management System using Segment Tree

Work Done Summary:

- Implemented a Segment Tree architecture in Java to manage range-based query analyses over regional hospital asset allocation tiers.
- Stored asset metrics systematically inside structural array buckets to keep aggregate tracking variables operational.
- Processed dynamic updates across discrete interval sectors as medical facilities adjusted resource pools or updated bed volumes.
- Executed range query operations to determine current resource counts or patient aggregates within selected facility tiers or geographical regions.
- Leveraged log-linear interval summation blocks to monitor data shifts and prevent equipment shortages across hospital units.

Implementation Details:

- **1. Configured SegmentTree array class** utilizing explicit storage arrays: `tree[]` and `resources[]` to capture continuous index groups.
- **2. buildTree()** constructs the initial internal balance values from foundational array logs in linear time.
- **3. update()** modifies resource availability counts at explicit indices, immediately bubbling up the delta corrections across parent ranges.

- **4. rangeQuery(left, right)** delivers aggregated inventory values across specified bounds in sub-linear logarithmic runtime.
- **5. Main class routine** triggers initial tree provisioning, executes inventory alterations, and extracts multiple dynamic interval results.

Result: Screenshots/Output

```
PS D:\JavaPrograms> javac HealthNetSegmentTree.java
PS D:\JavaPrograms> java HealthNetSegmentTree

Original Healthcare Sector Asset Inventory:
Sector 1 (Emergency Beds) = 120
Sector 2 (Ventilators) = 150
Sector 3 (Oxygen Concentrators) = 100
Sector 4 (Infusion Pumps) = 180
Sector 5 (Defibrillators) = 130

Calculated Segment Tree Range Array Structure:
Tree[1] = 680 (Total Consolidated Regional Assets)
Tree[2] = 370 | Tree[3] = 310

Executing Asset Range Query (Sector 2 to Sector 4):
Range Query Asset Sum Result = 430 available units

Updating Sector 3 Inventory (+70 new units delivered)...

Refreshed Range Query (Sector 2 to Sector 4) After Dynamic Update:
Updated Interval Asset Sum = 500 units

Data Audit Verification: Inventory balanced successfully.
```

Time Complexity & Empirical Analysis:

- Segment Tree Baseline Compilation: $O(n)$ initialization time.
- Dynamic Asset Adjustment: $O(\log n)$ bounded update paths.
- Range Query Evaluation: $O(\log n)$ search execution across any selected region.
- The Segment Tree delivers optimal support for heavy data metrics compared to baseline linear scan iterations.

GitHub Repository Link:

<https://github.com/bhavana-sri25/Sketch-DSA-CO2>

Student Signature

Faculty Signature